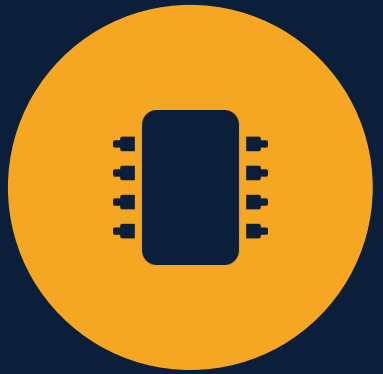
UNIT 3 · 9 hours

Zener Diode as a Voltage Regulator



Diodes, Transistors and their Applications

Basic Electrical and Electronics Engineering (EEE103)
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NMAM Institute of Technology, Nitte · 2025-26

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Learning objectives

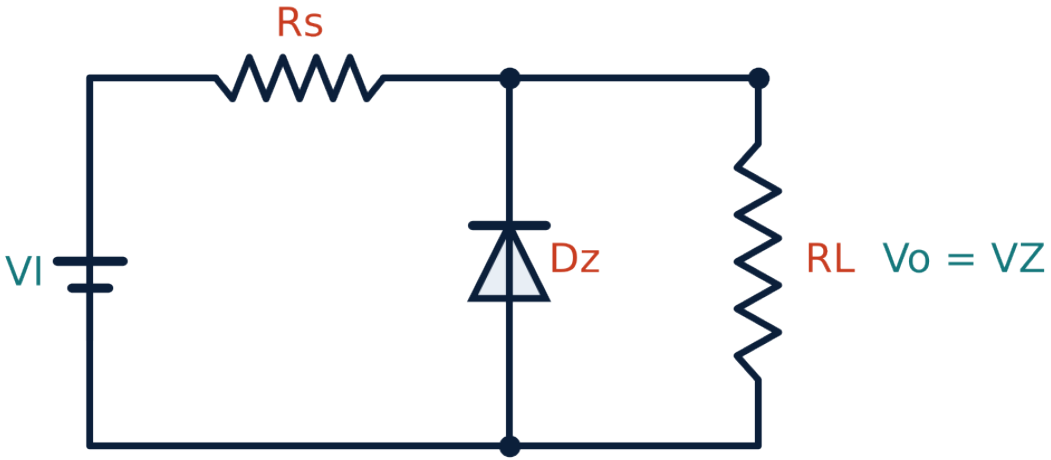
- Draw and explain the Zener shunt regulator
- Distinguish line and load regulation
- Design Rs for given limits
- Solve numericals

Syllabus context

Semiconductor diode, diode equivalent circuits, half-wave rectifier, full-wave bridge rectifier. Zener diode and its use in voltage regulation.

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Zener shunt regulator



- Zener connected in reverse across the load RL.
- Series resistor Rs limits current and drops the excess voltage.
- Vo = VZ as long as the Zener stays in breakdown.
- IS = IZ + IL; Rs = (VI − VZ)/IS.

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Types of regulation

Line regulation

Input voltage varies, RL fixed. Extra current flows through the Zener; Vo stays at VZ.

Load regulation

Input fixed, RL varies. Zener current changes to take up the difference; Vo stays at VZ.

Limits

Needs IZ ≥ IZmin always and IZ ≤ IZmax; so VImin and VImax and RL(min) are bounded.

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Design equations

$$RS = (VI(min) - VZ) / (IZmin + IL(max))$$

ensures Zener stays ON

$$IZ(max) = PZ / VZ$$

power limit

$$RS \geq (VI(max) - VZ) / (IZmax + IL(min))$$

protects the Zener

- Choose VZ = VL (required output).
- Power in Rs: PRS = IS^2 RS (choose a suitable wattage).

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Numerical 1 (from notes)

Problem A 5 V supply from 12 V DC using a 5 V, 2 W Zener. Find the maximum Zener current and the minimum series resistance.

- IZmax = PZ/VZ = 2/5 = 0.4 A
- Rs(min) = (VI − VZ)/IZmax = (12 − 5)/0.4

Answer: IZmax = 0.4 A, Rs(min) = 17.5 Ω

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Numerical 2: design a regulator

Problem Design a 5 V regulator from 12 V for a 100 Ω load. Choose IZ = 10 mA at full load.

- IL = 5/100 = 50 mA; IS = IZ + IL = 60 mA
- Rs = (12 − 5)/0.060 = 116.7 Ω → choose 120 Ω
- No load: IZ = 7/120 = 58 mA → PZ = 0.29 W: use a 0.5 W Zener
- PRS = IS^2 Rs = 0.06^2 × 120 = 0.43 W: use a 1 W resistor

Answer: Rs ≈ 120 Ω, 5 V / 0.5 W Zener, 1 W resistor

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Limitations

- Poor efficiency: power is wasted in Rs and the Zener.
- Output voltage fixed at VZ; limited load current.
- Ripple not fully removed: dynamic resistance rz.
- Better: IC regulators such as the 78XX series (Unit 5).

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Key takeaways

- Vo = VZ while the Zener is in breakdown
- IS = IZ + IL, Rs = (VI − VZ)/IS
- Line regulation: VI varies; load regulation: RL varies
- Check the no-load and full-load limits

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Practice questions

Q1. Explain with a circuit how a Zener regulates voltage.

Q2. A 12 V supply feeds a 6.2 V Zener, Rs = 100 Ω, RL = 200 Ω. Find IL, IS, IZ.

Q3. Find the Zener power if RL is removed.

Q4. Differentiate line regulation from load regulation.